

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) — Semester II Examination, 2023-24

Paper: GLB-201 — Elements of Mineralogy and Crystallography

Time: 3 Hours

Maximum Marks: 70

Ref. No.: 074430

*Instructions: Attempt **all** three Sections A, B, and C according to the directions.***SECTION — A**

*Answer **any two** questions. (14 marks each)*

Question 1. Why is the megascopic property of a mineral crucial for its nomenclature? A student described a mineral with the physical property “Perfect, 2 sets, {001}”. Which mineral property is he referring to? Explain in detail. [14]

Question 2. Define the optical properties of minerals and illustrate with diagrams the following properties (**any four**): [14]

- (a) Refractive index
- (b) Becke line test
- (c) Interference colour
- (d) Optical relief
- (e) Extinction angle
- (f) Pleochroism

Question 3. What is the basis for classifying silicate minerals? Discuss the different silicate groups and describe two processes through which quartz and feldspar can form. [14]

Question 4. Two olivine crystals (a common gemstone: “Peridot”) were excavated from basaltic rocks at two different depths within an igneous complex. Do they possess the same interfacial angles? Justify your answer with suitable reasons. Under which conditions is a reflecting goniometer used instead of a contact goniometer? Also explain the elements of symmetry in a three-dimensional coordinate reference frame. [14]

SECTION — B

*Answer **any two** questions. (14 marks each)*

Question 5. Write short notes on the following: [3.5 × 4 = 14]

- (a) What is the Nicol prism and why is it important in a polarizing microscope?

- (b) How to check the position of a polarizing microscope ready to examine the optical properties of a mineral.
- (c) What are isotropic and anisotropic minerals? How to identify these minerals under a microscope.
- (d) Define the optical properties that can distinguish quartz, plagioclase, microcline, and orthoclase with diagrams.

Question 6. Briefly explain the following sentences (**any four**) with suitable examples wherever necessary:

[3.5 × 4 = 14]

- (a) Pyrite generally breaks into a cubic crystal whereas calcite breaks into a rhombic crystal.
- (b) For some minerals, a streak plate reveals their streak, while for some, it indicates its hardness.
- (c) Quartz exhibits a range of colours, while plagioclase does not.
- (d) Anion or anionic group based classification of minerals is considered the most valid scheme.
- (e) Diamond and graphite, despite having the same chemical composition, have different specific gravity.

Question 7. Explain the following:

[7 × 2 = 14]

- (a) Why are 5-fold, 7-fold, and 8-fold rotational symmetries not possible in crystals?
- (b) In a cubic crystal, is the face (111) a general form or a special form?

SECTION — C

All questions are compulsory. (1 mark each)

Question 8. Answer in one/two words or sentence(s):

[1 × 14 = 14]

- (a) Write two optical properties that depend on the refractive index.
- (b) How to differentiate between cleavage and fracture under a microscope.
- (c) Write two essential mineral properties required to measure the extinction angle.
- (d) Define optical twinkling with an example of a twinkling mineral.
- (e) Write two optical properties to distinguish muscovite and biotite.
- (f) Write two different optical properties of hornblende and augite.
- (g) Which mineral is named after a scientist?
- (h) Who discovered the polarizing microscope, and in which year?
- (i) A mineral scratches topaz and shows good parting. Name the mineral.

- (j) Write one mineral formed through a metamorphic process showing a bladed habit.
- (k) Which instrument is used to distinguish pitchblende and hornblende?
- (l) Tiger's eye and cat's eye describe which property of a mineral?
- (m) What is a sphenoid?
- (n) What do you mean by a solid angle?